

Lie Groups and Lie Algebras

兹白大姨の数理研究

Author Be Like:



2026 年 7 月 28 日

University of Hamburg
Department of Physics

目录

1	Motivation: Lie Groups and Lie Algebras in Physics	1
1.1	Symmetry as invariance under a transformation	1
1.2	What exactly is a transformation?	3
1.3	From transformations to the abstract group concept	4
1.4	The Monster: how rich can a finite group be?	5
1.4.1	Several complementary descriptions	5
1.4.2	The first numerical glimpse of monstrous moonshine	5
1.5	Why study group theory?	9
1.5.1	Finite groups and classification	9
1.5.2	The Standard Model and continuous groups	9
1.5.3	Crystals and discrete groups	9
1.5.4	The Rubik's cube	10
1.6	Representations make abstract groups calculable	10
1.7	Transformations of quantum systems	11
1.8	Map of topics and physical applications	11
1.8.1	Background checklist from the lecture	12
1.9	Course organisation recorded in the source	13
2	Some Groups and Basic Properties	15
2.1	Order and first examples	15
2.2	Cayley tables, cancellation, and translations	17
2.2.1	Cayley tables	17
2.2.2	The cancellation rule	18
2.3	Abelian and cyclic groups	19
2.3.1	The unique group of order three	19
2.4	Groups of order four	21
2.4.1	The cyclic possibility	22
2.4.2	Completing all order-four Cayley tables	22
2.4.3	Several models of K_4	24
2.5	Subgroups and the order of an element	25
2.6	All subgroups of $\mathbb{Z}$	26

2.7	Presentations: generators and relations	27
2.8	The dihedral groups	29
2.8.1	The group D_3	29
2.8.2	The general group D_n	30
2.9	The quaternion group Q_8	31
3	The Symmetric Group and Cayley's Theorem	33
3.1	The symmetric group of a finite set	33
3.2	Left translations as permutations	34
3.3	The group G and the group L_G	36
3.4	Cayley's theorem	37
4	Cosets and Lagrange's Theorem	39
4.1	Left and right cosets	39
4.2	When are two cosets equal?	40
4.3	Congruence modulo a subgroup	41
4.4	Coset spaces and index	43
4.5	Lagrange's theorem	44
4.6	Complete coset calculations in D_3	45
4.6.1	The subgroup $H_1 = \langle a \rangle$	45
4.6.2	The subgroup $H_2 = \langle b \rangle$	46
5	Conjugation	48
5.1	Conjugation and conjugate elements	48
5.2	Conjugacy classes in $S_3 \cong D_3$	50
5.3	Centralizers	51
5.3.1	Centralizers in S_3	52
5.4	The size of a conjugacy class	53
5.5	The center of a group	55
5.6	The class equation	56
6	Homomorphisms	59
6.1	Definition and first properties	59
6.2	Images and preimages of subgroups	60
6.3	Image, kernel, and injectivity	61
6.4	Endomorphisms, isomorphisms, and automorphisms	62
6.5	Conjugation as an automorphism	64
6.6	Homomorphisms involving the integers	64
6.6.1	Multiplication by an integer	64
6.6.2	Reduction modulo m	65
6.7	Translations and the left regular representation	65

6.8	The sign homomorphism	66
6.9	The alternating group	68
6.10	Classification of cyclic groups	69
7	Normal Subgroups and Factor Groups	71
7.1	Why coset multiplication needs a condition	71
7.2	Normal subgroups	72
7.3	Normality is exactly what makes the product well defined	73
7.4	Factor groups	74
7.5	Kernels and images	74
7.6	The canonical projection	75
7.7	Simple groups	76
7.7.1	The classification picture for finite simple groups	76
7.7.2	Exactly which cyclic groups are simple?	78
7.8	The quotient S_n/A_n	79
8	The Symplectic Group	81
8.1	Hamiltonian mechanics in matrix form	81
8.1.1	Phase space	81
8.1.2	The standard symplectic matrix	82
8.2	Coordinate representations and canonical transformations	82
8.2.1	The same physical state in two coordinate systems	82
8.2.2	Deriving the Jacobian condition	83
8.3	Definition and elementary group properties	84
8.4	Canonical form of a nondegenerate skew matrix	85
8.5	All nondegenerate symplectic forms give isomorphic groups	86
8.6	Eigenvalues of a symplectic matrix	88
8.7	Examples of canonical transformations	89
8.7.1	The first restrictions on a point transformation	90
8.8	Poisson brackets	91
8.9	All canonical transformations with $q' = f(q)$	92
8.10	Reducing a general skew form to the standard form	94
8.11	Equations of motion as Poisson-bracket equations	94
8.12	Geometric discussion and symmetry groups	95
9	Direct Products of Groups	97
9.1	From a homomorphism to a quotient	97
9.2	The external direct product	98
9.2.1	Immediate properties	99
9.3	Presentations of direct products	101
9.4	The canonical factor subgroups	101

9.5	Recognizing an internal direct product	102
9.6	Products of finite cyclic groups	104
9.7	Further properties of direct products	106
9.7.1	Subgroups, normal subgroups, and quotients	106
9.7.2	Conjugacy classes and centralizers	107
9.7.3	Products of more than two groups	107
10	Matrix Groups	109
10.1	Fields, matrices, and characteristic	109
10.2	The general and special linear groups	109
10.2.1	The determinant homomorphism	110
10.2.2	The quotient isomorphism written directly	111
10.3	Orthogonal, unitary, and symplectic groups	111
10.4	Special orthogonal and special unitary groups	112
10.4.1	The orthogonal determinant	112
10.4.2	The unitary determinant	113
10.5	Scalar matrices and real general linear groups	113
10.5.1	Odd dimension	113
10.5.2	Even dimension	114
10.6	The relation between $U(n)$ and $SU(n) \times U(1)$	115
10.7	The complex general linear group	116
11	Gauge Invariance	119
11.1	The language of gauge theory	119
11.2	Minkowski spacetime and index notation	120
11.3	Maxwell theory in tensor form	120
11.3.1	Charge and current density	120
11.3.2	The electromagnetic field tensor	121
11.3.3	The inhomogeneous equations	121
11.3.4	The homogeneous equations	121
11.3.5	Charge conservation	121
11.4	Differential forms and the four-potential	122
11.4.1	The field strength as a two-form	122
11.4.2	Poincaré's lemma	123
11.5	The electromagnetic action	123
11.6	Gauge transformations and gauge freedom	124
11.6.1	The gauge group	125
11.7	Lorenz gauge and the wave equation	125
11.7.1	Why Lorenz gauge can be imposed	126
11.8	A nonrelativistic analogy	126

11.9	Noether's first theorem	127
11.10	Local gauge symmetry and Noether's second theorem	128
11.10.1	Why a constant gauge parameter is trivial for A_μ	128
11.10.2	Arbitrary local parameters produce an identity	128
12	Non-Abelian Gauge Groups and $SO(n)$ Yang–Mills Theory	130
12.1	A free multiplet of real scalar fields	130
12.1.1	Fields, spacetime, and the action	130
12.1.2	Lorentz invariance and the wave equations	131
12.2	The internal structure group $SO(n)$	132
12.2.1	The fundamental action on the multiplet	132
12.2.2	Infinitesimal rotations and the Lie algebra	133
12.3	Noether currents for the global internal symmetry	134
12.3.1	Direct check of infinitesimal invariance	134
12.3.2	The $\frac{1}{2}n(n-1)$ conserved currents	135
12.4	From a global symmetry to a gauge symmetry	135
12.4.1	Structure group versus gauge group	136
12.4.2	Why the ordinary derivative fails	137
12.5	The covariant derivative and the gauge field	137
12.5.1	Minimal substitution	137
12.5.2	Derivation of the connection transformation law	137
12.5.3	Why the transformed field remains Lie-algebra valued	138
12.6	Interactions forced by the gauge principle	139
12.6.1	Expansion of the matter Lagrangian	139
12.7	Dynamics of the gauge field	140
12.7.1	Why the Maxwell-type first guess is incomplete	140
12.7.2	Curvature as a commutator of covariant derivatives	140
12.7.3	Gauge covariance of the field strength	141
12.8	The Yang–Mills Lagrangian	141
12.8.1	Trace form and proof of gauge invariance	141
12.8.2	Self-interactions of non-Abelian gauge particles	142
12.9	The complete gauge-invariant theory	142
13	Lie Groups: Topology and Matrix Examples	144
13.1	Why continuous groups require topology	144
13.2	Topological spaces and continuous maps	145
13.2.1	Open sets and the axioms of a topology	145
13.2.2	Continuity as preservation of open preimages	146
13.3	Closed sets, closure, interior, and boundary	147
13.4	Covers and compactness	148

13.5	Connectedness and homeomorphisms	148
13.5.1	Connected and simply connected spaces	148
13.5.2	Homeomorphisms	149
13.6	Topological groups and basic examples	149
13.6.1	Subgroups and standard examples	150
13.6.2	Homomorphisms of topological groups	150
13.7	The identity component	151
13.8	The topology of the orthogonal group	152
13.9	The two components of $GL(n, \mathbb{R})$	153
13.10	Lie groups and linear Lie groups	155
13.10.1	General definition	155
13.10.2	Linear Lie groups and the closed subgroup theorem	155
13.11	Examples of linear Lie groups	156
13.11.1	Classical groups as automorphism groups of forms	156
13.11.2	The Borel and unitriangular groups	156
13.11.3	The affine group	157
14	Lie Algebras and Matrix Lie Algebras	159
14.1	From a Lie group to its infinitesimal algebra	159
14.2	Definition and immediate consequences	160
14.3	Generators and structure constants	160
14.4	First examples	161
14.4.1	Every one-dimensional Lie algebra is Abelian	161
14.4.2	$\mathbb{R}^3$ with the cross product	162
14.4.3	The general linear Lie algebra	162
14.5	Subalgebras and homomorphisms	163
14.5.1	$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3)$	164
14.6	Extracting a Lie algebra from a matrix Lie group	164
14.6.1	The orthogonal and special orthogonal algebras	165
14.6.2	Upper triangular matrices	165
14.7	The symplectic Lie algebra	165
14.7.1	Block form and dimension	166
14.7.2	The rank-one case and a correction	166
14.8	Unitary and special unitary Lie algebras	167
14.8.1	A basis and dimension for $\mathfrak{su}(2)$	168
14.8.2	Dimension of $\mathfrak{su}(n)$	168
14.9	The special linear algebra	168
14.10	Important groups and their Lie algebras	169
14.11	Brackets of subalgebras and the derived algebra	169
14.12	Structure constants, isomorphisms, and basis changes	170

14.12.1 Comparing two algebras in chosen bases	170
14.12.2 How the constants transform under a basis change	171
15 The Lie Correspondence and the Exponential Map	173
15.1 The matrix exponential	173
15.1.1 Definition and norm convergence	173
15.1.2 First algebraic properties	175
15.2 The local matrix logarithm	176
15.3 One-parameter subgroups	178
15.4 The Lie–Trotter product formula	180
15.4.1 Application: free energy and imaginary-time response	181
15.5 The Lie algebra of a closed matrix group	183
15.5.1 Exponentials lie in the identity component	185
15.6 Classical groups and their Lie algebras	185
15.6.1 General and special linear groups	186
15.6.2 Unitary and orthogonal groups	187
15.6.3 The physics convention	188
15.7 The exponential map of a linear Lie group	188
15.7.1 Local bijectivity on a closed matrix group	189
15.8 The tangent-space interpretation	190
15.9 Connected groups are generated by exponentials	192
15.10 When one exponential is enough	193
15.10.1 Surjectivity for $SU(n)$	193
15.10.2 Surjectivity for $SO(n)$	194
15.11 The identity component has the same Lie algebra	196
16 Derivations and the Adjoint Action	199
16.1 From operators to superoperators	199
16.2 Derivations of a Lie algebra	200
16.3 Inner derivations	201
16.4 The adjoint representation and the centre	203
16.5 The group adjoint action	204
16.6 Examples	206
17 Lie Group Homomorphisms and Their Differentials	208
17.1 Group and Lie-algebra homomorphisms	208
17.2 Differentiating a Lie group homomorphism	209
17.3 The adjoint homomorphism revisited	212
17.4 Hadamard’s lemma	213
17.5 Connected groups are determined infinitesimally	215
17.6 Continuity implies smoothness	215

17.7	Centres of connected groups and Lie algebras	216
18	The Double Cover $SU(2)$ to $SO(3)$	219
18.1	Pauli matrices and the real space $\mathfrak{su}(2)$	219
18.1.1	The normalized Frobenius inner product	220
18.2	Constructing the map by the adjoint action	221
18.2.1	The homomorphism property, index by index	221
18.2.2	Why $R(u)$ is orthogonal	222
18.2.3	Continuity	222
18.3	The induced Lie-algebra homomorphism	223
18.3.1	Cross-product matrices and the sign convention	223
18.4	The explicit rotation generated by u	224
18.4.1	Every element of $SU(2)$ is an exponential	224
18.4.2	Exponentiating the differential	225
18.4.3	The determinant is one	226
18.5	Surjectivity and the two-element kernel	226
18.5.1	Every rotation has a preimage	226
18.5.2	Computing the kernel from the exponential formula	227
18.6	Axis–angle coordinates and the topology of $SO(3)$	228
18.6.1	Closed paths through the identity	229
18.7	$SU(2)$ is the three-sphere	230
19	Covering Groups, Spin Groups, and Spinors	232
19.1	Covering homomorphisms	232
19.1.1	A discrete kernel is central when the cover is connected	233
19.2	First examples	234
19.2.1	The double cover of $SO(3)$	234
19.2.2	The exponential cover of the circle	234
19.3	The universal covering group	235
19.3.1	Every connected cover comes from the universal one	236
19.4	Pauli spinors in a magnetic field	236
19.4.1	The direction-dependent ground state	237
19.4.2	The spin expectation aligns with the field	237
19.5	Rotations of a spinor	238
19.5.1	A 2π rotation changes the spinor’s sign	239
19.6	Spin groups	240
19.6.1	Low-dimensional spin groups and their centres	240
19.7	Two further families of covers	241
19.7.1	The special unitary groups	241
19.7.2	The real symplectic group and the metaplectic group	242

19.8 The proper orthochronous Lorentz group	243
20 Linear Representations of Groups	245
20.1 From an abstract group to linear maps	245
20.1.1 Linear versus matrix representations	246
20.1.2 The multiplication law in indices	247
20.2 A first finite-group example: the quaternion group	247
20.3 Important adjectives for representations	248
20.4 Tensor representations of the rotation group	249
20.4.1 Scalars and vectors	249
20.4.2 Rank-two tensors	250
20.5 Linear response and Neumann's principle	250
20.5.1 Hooke's law and fourth-rank tensors	251
20.6 Infinite-dimensional representations on function spaces	254
20.6.1 The Helmholtz equation	254
20.6.2 Rotationally invariant Schrödinger operators	254
21 Equivalent Representations, Sums, and Products	257
21.1 Equivalent representations	257
21.1.1 Why this is an equivalence relation	258
21.2 Matrix representations and changes of basis	259
21.3 Unitarising a finite-group representation	260
21.4 Direct sums of vector spaces and linear maps	262
21.4.1 Sums of subspaces	262
21.4.2 Direct sums of maps	263
21.5 Direct sums of representations	264
21.6 Tensor products of vector spaces and maps	264
21.6.1 The tensor-product space	265
21.6.2 Tensor products of linear maps	265
21.7 Tensor products of representations	266
22 Invariant Subspaces and Irreducibility	269
22.1 Invariant subspaces and restricted representations	269
22.2 Reducibility and block-triangular form	270
22.2.1 An adapted basis	270
22.2.2 Example: rotations about one axis	271
22.3 Completely reducible representations	272
22.3.1 A reducible representation that is not completely reducible	273
22.4 Unitary representations and invariant complements	274
22.5 Maschke's theorem	275
22.5.1 The averaged-projection proof	276

22.5.2	A comparison of the hypotheses	277
22.6	Schur's lemma	278
22.7	The commutant and inequivalent irreducible summands	280
22.8	Irreducible representations of Abelian groups	281
22.9	Uniqueness of irreducible multiplicities	282
22.10	The defining representation of $GL(n, \mathbb{C})$	283
23	Symmetries and Degeneracies	285
23.1	Quantum transformations and Wigner's theorem	285
23.1.1	Two different uses of a unitary operator	286
23.2	Hamiltonian symmetry and the first examples	287
23.3	The symmetry-adapted decomposition	288
23.3.1	Constructing a common basis in the equivalent copies	289
23.3.2	The block-by-block commutator calculation	290
23.3.3	What symmetry does and does not determine	291
23.4	The spin- $\frac{1}{2}$ representation	292
23.5	Two spin- $\frac{1}{2}$ particles	294
23.5.1	Product and coupled bases	295
23.5.2	The spectrum without a 4 by 4 determinant	295
23.5.3	General spin- j matrices	296
23.6	Three and more spin- $\frac{1}{2}$ particles	297
23.6.1	Three spins and a nontrivial multiplicity matrix	297
23.6.2	Four spins and the coupling tree	299
23.6.3	The half-Pascal triangle and its closed formula	299
23.7	Tight-binding model on an open chain	300
23.8	Tight-binding model on a ring	303
23.8.1	Real Fourier subspaces	303
23.8.2	The complete $L = 6$ calculation	305
23.8.3	Irreducible decomposition and energy degeneracies	306